DSP Mathematics Reference Notes

Project Vibe-Synth | Internal Technical Document | v1.0 | 2026-05-10

This document collects the core signal-processing mathematics underpinning the algorithms generated by the Vibe-Synth DSP engine. It is a reference for contributors working on the generation layer, BIBO checker, and Faust template library. Equations use discrete-time notation throughout (n denotes the sample index).

1 Signal Basics and Notation

1.1 Discrete-time signals

A discrete-time signal x[n] is a sequence of real (or complex) numbers indexed by integer n. In Vibe-Synth, x[n] represents a single audio sample at time t = n / fs, where fs is the sample rate (default 44 100 Hz).

Unit impulse and unit step

delta[n] = 1 if n = 0, else 0

u[n] = 1 if n >= 0, else 0

Any FIR filter can be fully characterised by its impulse response h[n], the output when x[n] = delta[n].

1.2 Convolution

LTI systems produce their output via discrete convolution:

y[n] = (x * h)[n] = SUM_k x[k] * h[n - k]

For a length-M FIR filter this sum has exactly M non-zero terms.

2 Filter Design

2.1 First-order IIR low-pass filter

Difference equation:

y[n] = (1 - a) * x[n] + a * y[n-1]

Coefficient a for cutoff frequency fc:

a = exp(-2*pi * f_c / f_s)

BIBO stable for $|a| < 1$. In Faust:

```
lp1(fc) = _ : *(1 - a) : + ~ *(a)   with { a = exp(-2*ma.PI*fc/ma.SR); };
```

2.2 Resonant low-pass (biquad)

Second-order section difference equation:

$$y[n] = b0*x[n] + b1*x[n-1] + b2*x[n-2] - a1*y[n-1] - a2*y[n-2]$$

Coefficient	Formula
alpha	sin(w0)/(2*Q)
b0	(1-cos(w0))/2
b1	1-cos(w0)
b2	(1-cos(w0))/2
a0	1+alpha
a1	-2*cos(w0)
a2	1-alpha

Normalise all coefficients by a0. Stability requires $|a2| < 1$ and $|a1| < 1+a2$.

3 Reverberation Algorithms

3.1 Feedback comb filter

$$y[n] = x[n] + g * y[n - M]$$

M = delay length in samples; g = feedback gain. BIBO stability requires $|g| < 1$.

```
comb(M, g) = _ : (+ ~ (de.delay(8192, M) * g));
```

3.2 Schroeder reverberator

Combines parallel feedback combs with series all-pass filters. All-pass transfer function:

$$H(z) = (-g + z^{(-M)}) / (1 - g * z^{(-M)})$$

Unity magnitude for all frequencies. Freeverb (ef.reverb_freeverb) adds tuned comb delays and low-pass damping to simulate air absorption.

4 Waveshaping and Saturation

4.1 Cubic soft clipper

$$y = x - (x^3 / 3)$$

Taylor expansion of tanh(x) at order 3. Available as ef.cubicnl(drive, offset). Introduces only odd harmonics.

4.2 Hard clipping

y = clip(x, -A, A). Introduces odd and even harmonics. In Faust:

```
hardclip(A) = _ : min(A) : max(-A);
```

5 Amplitude and Frequency Modulation

5.1 Tremolo

$$y[n] = x[n] * (1 - d * (1 - (\sin(2*\pi*fLFO*n/fs) * 0.5 + 0.5)))$$

5.2 Chorus

$$tau(n) = tau\_base + D * \sin(2*\pi*fLFO*n/fs)$$

tau_base is typically 5-30 ms; D is the modulation depth in samples.

6 BIBO Stability Conditions

A system is BIBO stable if every bounded input produces a bounded output.

FIR filters are unconditionally BIBO stable.

IIR filters are BIBO stable iff all poles lie strictly inside the unit circle: |zk| < 1.

For y[n] = x[n] + g*y[n-M], the pole is at z = g^(1/M). Stability requires |g| < 1.

> BIBOChecker (compiler/bibo_checker.py) runs six heuristic checks: feedback gain >= 1.0, non-productive recursion, missing damping in reverb paths, high filter Q, excessive delay sizes, and large output gains.

7 z-Transform Quick Reference

The z-transform converts a discrete-time signal x[n] into X(z):

$$X(z) = SUM\_\{n=-inf\}\^\{inf\} \; x[n] * z\^(-n)$$

Operation / signal	z-Transform
x[n - k] (delay by k samples)	z^(-k) * X(z)
x[n] + y[n]	X(z) + Y(z)
a * x[n]	a * X(z)

delta[n] (unit impulse)	1
u[n] (unit step)	z / (z - 1)
a^n * u[n]	z / (z - a), |z| > |a|

7.1 Transfer function and frequency response

For an LTI system, H(z) = Y(z)/X(z). Frequency response is H(z) evaluated on the unit circle:

H(e^{j*omega}) = H(z)|_{z = e^{j*omega}}

omega = 2*pi*f/fs is the normalised angular frequency (0 to pi for 0 to fs/2).

8 Glossary

BIBO — Bounded-Input Bounded-Output — a stability criterion for LTI systems.

Biquad — A second-order IIR filter section (two poles, two zeros).

Comb filter — A filter whose frequency response resembles the teeth of a comb, produced by adding a delayed copy of the signal.

Difference equation — A recurrence relation expressing the current output sample as a linear combination of past inputs and past outputs.

Faust — A functional programming language for real-time audio DSP that compiles to C++, WebAssembly, and other targets.

FIR — Finite Impulse Response — output depends only on current and past inputs. Always BIBO stable.

IIR — Infinite Impulse Response — output depends on past outputs as well as inputs. Stable iff all poles inside the unit circle.

LFO — Low-Frequency Oscillator — a sine or triangle wave below ~20 Hz used for modulation.

LTI — Linear Time-Invariant — fully characterised by its impulse response.

Pole — A value of z where H(z) goes to infinity. Poles inside the unit circle correspond to stable decaying exponentials.

Q factor — Quality factor — determines the bandwidth of a resonant filter peak. Higher Q = narrower peak.

Waveshaper — A memoryless nonlinear function applied to each sample to introduce harmonic distortion.

2026/05/12 Note： Update this document when new DSP archetypes are added to dsp_templates.json.